11

The social construction of psychological knowledge

The preceding analysis of the historical development of investigative practices in psychology is obviously based on a certain model of the investigative process. Although various aspects of this model have emerged in previous chapters, some of the basic features of the model still need to be made explicit. The first part of this chapter is devoted to this task. In the remainder of the chapter I briefly address some general questions pertaining to the status of psychological knowledge that arise out of the kind of analysis presented here. These questions all revolve around the issue of how the social construction of psychological knowledge affects the reach of the knowledge product. Although knowledge claims are generated in specific sociohistorical situations, it does not follow that they have no significance or validity beyond those situations. There are two possibilities here. First, psychological knowledge may be generalizable to situations other than those in which it was generated. This is the question of applicability. More fundamentally, however, there is the possibility that psychological knowledge claims may be able to tap a level of "psychological reality" that is independent of the special conditions under which it is investigated. The second and third sections of this chapter discuss these possibilities.

The political economy of knowledge production

Investigative practices are part of a productive work process. Their employment results in the generation of valued products, though of course these are not primarily material but symbolic products – namely, scientific knowledge claims. This same employment necessarily involves people who put these practices to work and also some kind of raw material that is productively transformed into the valued knowledge product. Producers of scientific knowledge never work as independent individuals but are enmeshed in a network of social relationships.[1] What they initially produce is not so much knowledge as knowledge claims. Such claims are only transformed into knowledge by an acceptance process that involves a number of individuals – such as reviewers, readers, textbook writers – who share certain norms and interests.[2] The anticipation of this acceptance process affects the production of knowledge from the beginning.

Like all human productive activities the production of scientific knowledge is highly goal-directed. This directive component manifests itself not only in the fact that so much scientific activity is explicitly devoted to hypothesis testing, but also in the more implicit commitment to the search for a certain type of knowledge. The investigator is not just interested in any kind of answer to specific questions; he or she is only interested in obtaining such answers on certain terms that have been set in advance. Knowledge products must take a particular form: They may have to be quantitative, for example; they may be limited to intersubjectively observable phenomena; they may be required to divide the world into precisely defined variables. These formal prescriptions, which dictate in advance what form the knowledge product shall take, often become quite specific. For example, they may dictate that only linear and additive relations among variables shall appear in the product, or that the product shall take the form of interindividual differences. As we have seen in this volume, the kind of knowledge product desired has a decisive influence on the choice of investigative procedures.

Prescriptions pertaining to the form of knowledge products are themselves social products. They are characteristic of particular groups of knowledge producers and unite their members in common methodological commitments. More fundamentally, they tend to be products of the interface between the group of knowledge producers and centers of social power that are able to determine the success or failure of their enterprise.[3] Thus, the investigators gathered in the seventeenth-century Royal Society decided to pursue a form of knowledge that promised technical utility and to exclude forms of knowledge that were tied up with questions of social reform and revolution. The end result was the modern conception of the "value neutrality" of scientific knowledge.[4] But the decision to limit the forms of admissible knowledge was made because producers of knowledge lived and worked in a social context where certain types of knowledge were perceived as useful and others as threatening to powerful social interests.

In other words, rules about the admissibility and desirability of particular kinds of knowledge product are generated in the course of a *historical* process of negotiation that involves the coming to terms of knowledge producers with the realities of social power and influence in the world they inhabit. Direct relationships with the most important centers of political and economic power are sometimes involved, although in complex industrial societies this power is increasingly mediated by secondary groups that exercise power in limited areas. Any new group of knowledge producers, like scientific psychologists, will have to come to terms with a world that already contains knowledge producers that make related claims, such as medical professionals, educationists, and experimental physiologists. These established groups wield a certain amount of social power, and that power is based on their institutionalized monopoly over certain types of knowledge product. How can a new group of knowledge producers thrive under these conditions?

It can thrive only if it manages to form effective alliances. This it can do by enlisting the interest of established groups in its knowledge products and avoiding their censure, a process that has many facets.[5] First of all, the new knowledge products had better reflect well-established preconceptions about the forms of valuable knowledge. If quantitative knowledge is particularly valued, then it helps to establish new claims if one can give them a quantitative form. But established preconceptions about the form of knowledge products automatically extend to the methods used to generate them, for the form of the product depends directly on the nature of the method of production. So one must be seen to be engaging in practices that produce the right type of knowledge, even though such rituals have more in common with magic than with science.[6] More specifically, the borrowing of investigative practices from better-established fields, like experimental physiology, provides a basis for limited alliances that link the new field and its products to the existing network of recognized scientific knowledge.

But this kind of alliance would provide a very limited basis for autonomous development if the new knowledge products were not seen as having a significant social value of their own. They must become marketable, and that means that there must be categories of persons to whose interests the new product is able to appeal. The more powerful and better organized these consumers of knowledge products are, the more successful the producers will be in consolidating their own position. American psychologists scored some real successes in this direction by providing knowledge products that mobilized the interests of educational and military administrators as well as the administrators of private foundations.[7] Even if the alliances so formed were often temporary, and based more on promissory notes than on real goods, they served an important function in establishing the credentials of the new discipline at a critical stage in its development.[8]

The relatively new knowledge products that scientific psychology had to offer were hardly dumped on an empty landscape but had to be carefully inserted into a terrain whose commanding positions were already occupied. There were important national differences in regard to who occupied the commanding positions and how firm the occupation was, as we saw in chapter 8. But the crucial point is that the successful establishment of a new discipline is very much a *political* process in which alliances have to be formed, competitors have to be defeated, programs have to be formulated, recruits have to be won, power bases have to be captured, organizations have to be formed, and so on. These political exigencies necessarily leave their mark on the discipline itself, and not least on its investigative practices. The political environment largely determines what types of knowledge product can be successfully marketed at a particular time and place. The goal of producing an appropriate type of product plays a crucial role in the selection of investigative practices within the discipline. Ultimately, the limitation of investigative practices results in corresponding limitations on the knowledge products that the discipline has to offer. So in the end, those with sufficient social power to have an input into this process are likely to get the kinds of knowledge products that are compatible with their interests.

Being obliged to operate in an external political environment, the community of knowledge producers is bound to develop an internal political environment. In the face of the external world, this community constitutes a special-interest group, but these special interests presuppose a more general system of knowledge production within which they are recognizable as special interests. There is an intimate relationship between the general forms of presuppositions, knowledge goals, and investigative practices and their specific embodiment.[9] As the community of knowledge producers grows it develops internal norms and values that reflect its external alliances. Its professional project is directed at carving out and filling a particular set of niches in the professional ecosystem of its society, and its internal norms reflect the conditions for the success of this project.[10] These norms tend to govern both the production of knowledge and the production of the producers of knowledge through appropriate training programs.

Individual research workers tend to take these norms for granted and therefore believe they are making rational and technical decisions about the methodology they adopt. They forget that the historical development of the discipline has preselected the kinds of alternatives realistically available to them. In the case of psychology, it is particularly implausible to maintain that this preselection was rationally determined. Given their historical situation, individual investigators may well be making rationally defensible choices about their research practice, but there is no guarantee that the sum of these choices somehow constitutes a rational development of the discipline as a whole. The history of twentieth-century psychology provides little comfort for believers in the existence of a rational "hidden hand" that steers development on the disciplinary level. If one accepts sharp limitations of time and place, and if one stays within a very restricted framework of assumptions, one can make a case for some spurts of more or less rational development. But if one adopts a broader perspective, the political fundamentals become impossible to ignore.

This is also the case when one examines the investigative situations set up in the course of disciplinary practice. These have a political dimension because they necessarily involve questions of rights and questions of power. In the type of investigative practice that came to dominate the discipline, investigators are clearly in a position of power and authority vis-à-vis their human subjects. They give the directions, they manipulate the situation, they control the setting, it is their purposes that determine the course of events, and so on. But there are limits to what they can get away with, because ultimately the human and civil rights of experimental subjects must be recognized if unpleasant political consequences for the profession and legal consequences for its individual members are to be avoided. In the language of professional ideology, the rights of subjects come under the heading of research ethics while the powers of investigators come under the heading of technique. But of course both are part of the political aspect of the research process and are closely related.[11]

Investigative practices involving human subjects have always involved implicit political decisions (see chapter 4). Once a certain investigative style had established its hegemony, the question of subjects' rights became largely restricted to negative freedoms, such as limits on what could be done to subjects or limits on permissible degrees of ignorance about what was going on. But, as we have seen, in an earlier period the positive rights of subjects were an important issue. In particular, there was the question of whether the experience of subjects was to count within the framework of the investigative situation. The Wundtian experiment, the psychoanalytic situation, the Gestalt and Lewinian experiments all answered this question in the affirmative. However, it became a key principle of the dominant model of psychological experimentation that the subject's experience was to be discounted.[12] Attempts to change this state of affairs have always evolved the most determined resistance. At one time behaviorist dogma served a purpose in justifying what amounted to the disenfranchisement of experimental subjects. When this dogma became discredited, other reasons were found. The resistance is not surprising because taking subjects' experiences seriously strikes at the basic political structure of the dominant form of psychological experimentation.

Political questions regarding participants in investigative situations are not a peculiarity of psychology. From the beginning of modern scientific experimentation in the seventeenth century, the question had to be faced of how agreement was to be reached on basic facts. The solution that won the day involved reliance on the testimony of reliable witnesses about what they had observed, and the reliability of witnesses depended on their social status.[13] This was clearly a political decision, which confirmed that the pursuit of science was to be limited to an elite. Psychologists had to face a similar decision, although in their case the consequences were probably more profound. One could do as the physical sciences had done and limit participation in experiments to a community of elite observers whose experiences provided the basic information on which the science relied. This pattern was followed in the Leipzig experiments and by various successors. Alternatively, one could admit anyone to participation in psychological investigation and then establish a kind of class system within the investigative situation, so that only the investigators counted as competent witnesses while their subjects were limited to the role of objects of investigation. This alternative, as we know, proved to be far more successful than the first.

The most important point, however, is that the political forms imposed on the knowledge-generating situations profoundly affected the kind of knowledge that was produced. Whereas in the physical sciences the social arrangements among experimental participants did not in themselves alter the material to be investigated, this was hardly the case in psychology. Here, the object of investigation was itself part of the social context of investigation. The persons, whose "responses" formed the basis for the psychological knowledge product, were themselves part of the social context of investigation. This entailed a rather intimate link between knowledge product and investigative situation, a link that could pose a threat to the generalizability of psychological knowledge. To what extent could such knowledge be applied in social situations that differed from those in which it was produced? This question requires some consideration within the present framework.

Expert knowledge and life outside the laboratory

The relationship of the producers of scientific psychological knowledge to other knowledge producers is not only a question of interprofessional alliance and competition. Before such questions can become significant in the development of a field, it must achieve a minimal status as a science. It must have earned sufficient respect as a knowledge-producing enterprise to be a serious candidate for competition or for alliance with other professionalized fields. That means its products must have become clearly distinguished from the everyday or common knowledge and belief of the lay public and achieved the status of *expert* knowledge.[14] There is nothing more inimical to a field's success as a source of valued knowledge than the suspicion that it is able to supply no more than a duplication of what "everyone" knows anyway or, worse, a reinforcement of popular superstitions. The "soft" areas of psychology and parapsychology have always had to contend with this problem.

There had to be constant vigilance about maintaining the sharpest possible differentiation from folk knowledge. This was not a problem limited to some marginal areas but affected the discipline as a whole. In the judgment of the lay public everyone had to rely on psychological knowledge in making his or her way through the world. How could anything offered by experts compete with a lifetime of experience in human affairs? This ever-present background challenge had effects on the investigative practices of psychologists that should not be underestimated. Whatever else they may have done, these practices also served to demonstrate a crucial *distance* from those mundane situations in which everyday psychological knowledge was acquired. This was achieved largely by drawing on the mystique of the laboratory and the mystique of numbers, both of which had been well established prior to the appearance of modern psychology. The very artificiality of laboratory situations became a plus in establishing the credentials of knowledge claims emanating from this source, and the imposition of a numerical form on otherwise trivial knowledge gave it an apparent significance with which lay knowledge could not compete. Replacing ordinary language with jargon helped too.[15]

Nevertheless, the relationship between expert and everyday knowledge had other aspects. For one thing, investigative practices involving human subjects were unavoidably social practices and as such were not as clearly distinct from extralaboratory practices as the experimental mystique made out. In order to generate psychological knowledge, if one was not to rely solely on self-observation or on animal analogies, one had to set up social situations involving clearly patterned relationships among the human participants. These situations resembled certain more familiar social situations, not only because they were born in already established institutional environments, but also because the type of knowledge product desired had affinities with other knowledge products gathered in those environments. As we saw in chapter 4, clinical experiments emerged in a clinical environment that had preformed the social relationships among investigators and their "subjects" as well as the shape of the knowledge product that resulted from their interaction. Similarly, mental testing mimicked the social situation of a school examination in a recognizable way, and its product, the objective grading of individual performances, represented the optimal result of an idealized examination. In the last analysis psychological investigative situations constituted a development of already existing social practices.[16]

This was just as well, for the relevance of psychological knowledge products to life outside the research setting depended on this link. Clinical knowledge was applicable in clinical situations, and the knowledge gained by means of mental tests was applicable in settings devoted to the grading and selection of individuals on strict performance criteria. In these cases the homology of the investigative and the application setting made possible a direct transfer of information from the one to the other. Where this homology did not exist, the general relevance of research products required an elaborate apparatus of speculation, which made the whole enterprise an easy target for criticism. Often the viability of generalizations from research depended on the effectiveness of certain ideological presuppositions. Thus, whereas the application of results from intelligence tests to selection problems in schools was no more problematic than the use of any other kind of examination result, more general inferences about human intelligence were bound to be far more controversial. They depended on elaborate assumptions about the structure of the human mind and about the role of heredity in human affairs. Generalizations from research in other areas also depended on the plausibility of certain fundamental assumptions.[17]

One assumption was so pervasive in generalizations from psychological research that it deserves special mention. The nature of this assumption is revealed by the way in which the regularities emerging out of investigative situations were reattributed. They were attributed, of course, to the individuals who participated in the investigation as subjects. However, the regularities in question were not taken as conveying information about an individual-in-a-situation but about an individual in isolation whose characteristics existed independently of any social involvement. This Robinson Crusoe assumption played an increasingly important role the more psychology turned from psychophysiological investigations to generalizations about human conduct on the molar level. Faith in the Robinson Crusoe myth made it seem eminently reasonable to ignore the settings that had produced the human behavior to be studied and to reattribute it as a property of individuals-in-isolation.

Psychologists were not the inventors of this myth.[18] Their discipline was a relatively late product of an individualistic civilization that had long cherished the ideal of the independent individual for whose encapsulated qualities all social relations were external. In adopting this myth as a fundamental and unquestioned assumption for their work, psychologists were showing themselves to be no more than good citizens of the society that nourished them. Yet their role was not entirely passive. For by taking the myth for granted in a context that was supposed to be devoted to objective knowledge, they were in effect giving it a new justification and legitimacy. Their procedures for interpreting the products of their investigations were a kind of enactment of the founding myth that they shared with their compatriots. Moreover, psychological research often incorporated individualistic presuppositions in a uniquely unadulterated and often crass manner. More than the contributions of other social scientists, the work of psychologists represented a kind of celebration of the myth of the independent individual in its pure form.[19]

Scientific psychology found itself in this position, not only because of the nature of the subject matter that had been bequeathed to it from its prescientific phase, but also because the natural scientific methodology, on which it staked its prestige, converged nicely with its traditional assumptions. Experimental methods isolated individuals from the social context of their existence and sought to establish timeless laws of individual behavior by analogy with the laws of natural science. Shared social meanings and relations were automatically broken up into the properties of separate individuals and features of an environment that was external to each of them. The functional relationships established in this way presupposed an individualistic model of human existence.[20] Anything social became a matter of external influence that did not affect the identity of the individual under study.

This division of the actions of socially situated persons into separate individual and external situational components provided the framework for a general atomization of the features of both the individual and the situation. The latter had become an "environment" for the independent individual. A pursuit of universally valid natural scientific laws most often took the form of establishing empirical associations between separate individual and environmental "variables" and "factors." This enterprise depended on the crucial assumption that the variables studied would retain their identity irrespective of context.[21] Discounting of context applied both on the individual and on the situational level. Only by assuming that the responses of several individuals to a psychological instrument were manifestations of an identical psychological property that remained qualitatively the same, irrespective of differing personal contexts, did it make sense to combine individual responses into the kind of abstract collectivity that was discussed in chapter 5. Similarly, it only made sense to develop theories about psychological features like learning in the abstract, anxiety in the abstract, or intelligence in the abstract, if one assumed that such features remained identical irrespective of the situations in which they were investigated.[22]

Applications of psychological research products tended to fall along a scale from specific applications in mundane situations to grand claims about human nature that had profound implications for social policy. Among the latter one would find claims about the existence and heritability of "general intelligence" or claims about the origins and social role of race prejudice. At the other end of the scale there would be local decisions about individuals on the basis of test results, or improvements in human engineering. Now, it is easy to see that the two kinds of application differ in terms of the distance between the investigative situation and the situation to which the research products are to be applied. In the case of grand applications to social policy there is a vast distance between the situations in which the research products are generated and the kinds of situation to which they are to be applied. This distance has to be bridged by a host of unproven and often unspoken assumptions, and thus the whole enterprise is essentially based on leaps of faith. But in the case of narrow, technical applications of psychological knowledge there is typically a considerable degree of continuity between the investigative situation and the situation in which the application takes place. When one applies intelligence- or aptitude-test results to the prediction of future performance in the appropriate settings, academic or otherwise, one is essentially using a *simulation* technique. The more effectively the investigative context simulates the context of application, the better the prediction will be.[23]

During the period under review in this volume such a formulation would not have been acceptable, or perhaps even comprehensible, to most psychologists because of their metaphysical commitment to the reality of a world made up of fundamentally independent individuals. With such a starting point there were two kinds of approach that psychologists could adopt toward their subject matter. They could either set about studying concrete individuals and their specific attributes, or they could study abstract relationships that held universally across individuals. The former alternative was incompatible with conventional notions of scientificity, so the second alternative carried the day. Eventually, this long-recognized alternative was named and codified in terms of the well-known distinction between the idiographic and nomothetic approach.[24] As understood by most psychologists, this distinction generally served to rationalize their search for universal "laws" or principles. This was because the division of methods into idiographic and nomothetic was presented as though it exhausted the possibilities, and because the idiographic alternative was given a strictly individualistic interpretation. Faced with the choice between the study of individual biography and the study of "scientific laws of human behavior," most psychologists had no difficulty in deciding on the more ambitious alternative.

Unfortunately, this set of false alternatives removed the most promising possibilities from view. Those possibilities open up with a recognition of the fact that a series of situations, including investigative contexts, can have structural features in common. Individual behavior, being always part of such situations, would then show common features across a range of structurally similar situations. This would make generalization and prediction from one context to another possible, although such generalizations would always have to be regarded as conditional on the comparability of situations in terms of the similarity of their crucial structural features.

However, we should note that there are two ways in which the distance between the context of investigation and the context of application can be narrowed. Letting the context of investigation simulate the context of application is one way, and certainly the more easily recognized way. But in many cases of successful application of psychological knowledge one can also detect a reciprocal process that involves a change in the context of application so that it comes to resemble the context of investigation. The conventional model of the application of scientific knowledge involves a fixed context of application, but of course this is not true in the real world. Rather, the application involves an actual change of practices, which are transferred from the artificial laboratory setting to the world outside. In other words, the application of knowledge is possible only insofar as an artificial construction, derived from the investigative situation, is imposed on the natural world. This is what happens in the construction of industrial plants or in the adoption of sterile conditions in medicine, for example.[25]

An analogous process operates in the case of psychological knowledge. Applying mental tests often meant modifying some of the practices of schools so that they resembled testing practices more closely. Having spawned intelligence tests in the first place, some school examinations subsequently imitated their offspring in terms of such matters as timing. question format, and use of statistical norms. Or, to take another example from a later period, the adoption of behavior-modification programs in institutions involved the restructuring of the context of application so that it more closely resembled a particular investigative context. Practices have always wandered from noninvestigative to investigative contexts and back again. Genuine application of psychological knowledge depends on this, for the bond between knowledge and the practices with which it is associated is an extremely intimate one. Abstract knowledge only exists abstractly; its application requires a transfer of corresponding practices. So if the purely ideological application of psychological knowledge in support of particular social policies were ever to be converted into a real application that affected its objects directly, it would require an appropriate reconstruction of society, as B. F. Skinner so clearly demonstrated in *Walden Two*.[26]

It is apparent that the question of the distance that separated investigative situations from the situations in which their knowledge products were applied involved psychologists in a serious contradiction. On the one hand, the rhetoric of science required that this distance be emphasized and magnified. Because the yield from investigative situations was supposed to consist of universally valid generalizations (so-called nomothetic laws), these situations were endowed with a mystique that rendered them so remote from ordinary life that they were not even seen as social situations. Even the idea that there might be a social psychology of psychological experiments only arose at a late stage in the development of the discipline.[27] However, there remained the rather indigestible fact that the discipline's ability to make fairly reliable predictions about human beings outside the psychological laboratory depended to a large extent on the closeness of the context of investigation to the context of application. It was often the case that psychological knowledge had some technical utility only insofar as its investigative practices were continuous with relevant social practices outside the investigative situation.

This contradiction between scientistic rhetoric and the facts of life in applied psychology tended to maintain the separation of "pure" and applied research. Politically, both the rhetoric of science and specific technical utility were indispensable for the rapid development of the discipline and both continued to flourish side by side. But the political effect was enhanced by a professional ideology that interpreted any successes of so-called applied research in terms of the "application" of knowledge derived from "pure" research.[28] In this way each partner could profit from the value attributed to the activities of the other (see chapter 8). "Pure" research could claim support on the basis of its ultimate practical usefulness and "applied" research could speak more authoritatively by clothing itself in the mantle of science. In actual fact, "applied" research usually relied on its own practices with little or no help from "pure" research, and "pure" research either continued in complete isolation or adapted the practices of "applied" research, contributing little but technical sophistication and a more abstract terminology. Throughout the period under review here, the major function of laboratory research was probably the professional socialization of aspirant members of the community of producers of psychological knowledge. Laboratory work imparted a very specific interpretation of the meaning of "science," and one must suspect that most of the products of this work existed mainly to fill the pages of textbooks.

Below the rhetoric of pure and applied science it is possible to discern a developing fundamental convergence between contexts of investigation and contexts of application. In the investigative contexts that became increasingly popular during the first half of the twentieth century, the individuals under investigation became the objects for the exercise of a certain kind of social power. This was not a personal, let alone violent, kind of power, but the kind of impersonal power that Foucault has characterized as being based on "discipline."[29] It is the kind of power that is involved in the management of persons through the subjection of individual action to an imposed analytic framework and cumulative measures of performance. The quantitative comparison and evaluation of these evoked individual performances then leads to an ordering of individuals under statistical norms. Such procedures are at the same time techniques for disciplining individuals and the basis of methods for producing a certain kind of knowledge. As disciplinary techniques the relevant practices had arisen during the historical transformation of certain social institutions, like schools, hospitals, military institutions, and, one may add, industrial and commercial institutions.[30] Eventually, the knowledge-generating potential of these kinds of practices became realized in an increasingly systematic way, and the knowledge so produced was fed back into the original disciplinary institutions to increase their efficiency. This kind of knowledge was essentially administratively useful knowledge required to rationalize techniques of social control in certain institutional contexts. Insofar as it had become devoted to the production of such knowledge, mid-twentieth-century psychology had been transformed into an administrative science.

The limits of constructivism

The profound dependence of the psychological knowledge product on the nature of the practices that generated it did not hurt the "external validity" of the product.[31] As we have suggested, that depended on the similarity of the context of investigation and the context of application. Because the common practices of investigation had arisen by an extension of previously existing social practices there were always certain areas of life that had already been prestructured for the insertion of a certain kind of psychological knowledge. The fact that psychological knowledge was constructed, rather than discovered by naive inspection, did not mean that it had no general validity. As long as the principles used in the construction of the knowledge product were not totally unique, the possibility of at least a degree of generalization beyond the investigative situation was always present. However, this possibility was always circumscribed by the extension of the relevant social conditions.

Psychology of course aimed at a different kind of generality for its knowledge products. Large parts of it, at any rate, pretended to a knowledge that was universal, ahistorical, and independent of both the context of investigation and the context of application. Such knowledge almost always turned out to be chimerical, although even in this form it may have had an important social function. It may have provided a pseudoscientific affirmation of various ideological positions that had been presupposed in the construction of the knowledge in question, particularly an ideology of abstract individualism and an ideology of social control.[32] On this level generalizations from psychological research also referred to a certain reality outside the research situation. Although it was only the reality of an image of human nature and society, faith in such images could have material social consequences. In the long run, the contribution made to this kind of faith may have represented a more successful application of psychological knowledge than any technical application in specific institutional contexts.[33]

Neither the technical nor the ideological validity of psychological knowledge provided it with more than local significance. Technically, its relevance depended on the existence of a certain institutional framework, and ideologically its plausibility extended only as far as the cultural forms in which the shared faith was expressed. The question that remains is whether the investigative practices of psychology could ever yield a kind of knowledge whose significance was more than local. In this volume we have been examining the dependence of the knowledge product on the conditions of its production, and this has necessarily entailed a deconstruction of the generally false claims to universality that were commonly made on behalf of psychological knowledge. Does this mean that such claims were always and necessarily false? When allowance is made for the factors that led to a relativizing of psychological knowledge, is there no remainder?

Let us be clear that this question is not concerned with knowledge about the physiological conditions of human action and experience but with specifically psychological knowledge. A relatively small proportion of research within the discipline of psychology has always taken these physiological conditions and their effects as its subject matter. But psychology became an autonomous discipline only when such investigations were banished to a corner of the field and attention was concentrated on purely psychological knowledge. The question is whether the latter ever managed to have reference to a reality that was not simply a product of the investigative context.

Such a reality would have to exist on some level other than that of the regularities established within specific investigative situations. For these regularities, we have seen, are largely a product of various investigative practices. If the history of psychological research demonstrates anything, it demonstrates the extraordinary pliability of human beings. Of course, there are limits to this pliability. There are obvious physiological limits, though these are not particularly problematical. Psychological limits that are themselves a product of the investigative context, as when an experimental task is too fatiguing or boring or confusing, can easily be accommodated by a constructivist paradigm, as can those limits that are a function of the particular social background of experimental subjects. But are there general *psychological* sources for the limitations on human pliability in investigative situations? Such sources would constitute a level of psychological reality that is not to be confused with the overt regularities observed in the course of investigation.

As long as psychological reality is identified with the overt regularities of behavior established in investigative situations, there can be no escape from the consequences of a purely constructivist perspective. Such regularities are undoubtedly a product of social construction. However, constructivism may not have the last word if we recognize the fundamental distinction between what, in Bhaskar's realist philosophy of science, is called the domain of the real and the domain of the actual.[34] The latter comprises the observable events that, under specific conditions, can form the basis for empirical domains. The domain of the real, however, involves generative mechanisms that are not observable in themselves and that exist independently of any investigative intervention. Scientific research, whether in psychology or in physics, aims to increase our knowledge of such generative mechanisms.[35] Observed regularities are only of interest insofar as they assist in this task.

In the earlier phases of the history of modern psychology this kind of perspective was not uncommon. Wundt himself introduced the concept of *psychic causality,* which covered the psychological generative principles that were at work both in psychological experiments and in everyday life.[36]

Psychic causality made the difference between organismic responding and the construction of an organized subjective world. Although Wundt declared the exploration of psychic causality to be the aim of psychological research, there was a huge gap between such aims and his methods, so that his conception of psychic causality remained largely a matter of good intentions. Nevertheless, he had boldly confronted a fundamental issue which most of his successors preferred to sidestep.

Another psychologist who recognized the crucial importance of psychic causality, or psychic determinism as he called it, was Freud. He consistently tried to explain his clinical observations, as well as certain patterns taken from "everyday life," in terms of an underlying level of psychological causal processes. His version of psychological causality was very different in content from that of Wundt, but it was also far more specific. Nevertheless, the relationship between the level of empirical observation and the level of generative mechanisms remained problematic.

Gestalt psychology and the related Lewinian psychology also depended on notions of psychological causality that now took the form of models for the operation of forces in psychological fields.[37] These models were always underdetermined by the empirical evidence; they referred to a domain of causes that was different from the domain of observations. Toward the end of our period Jean Piaget began to develop yet another approach to the problem of psychological causality but his model did not become widely known until much later.

These and other less prominent attempts at coming to grips with the issue of psychological causality foundered on two interdependent problems. The various conceptions of the nature of psychological causality were clearly widely divergent and their ties to their respective empirical domains were insecure. Neither of these problems was peculiar to psychology. Fundamental differences about the nature of physical causality were far from being an unknown phenomenon in the history of physics, and even the underdetermination of theoretical models by empirical domains had never in itself been a cause for panic. However, in the historical context of a discipline desperate to establish its status as a reliable source of certain and practically useful knowledge, these fundamental dissensions were easily seen as threats. The ascendancy of positivist philosophies of science during the critical formative period of the discipline also meant that its struggle for legitimacy had to be fought in an intellectual environment that was hostile to conceptions of reality that went beyond the phenomenal level.[38]

It became common to identify psychological reality with empirical regularities. The gradual adoption of the rule that all theoretical concepts had to be "operationally defined," and the peculiarly narrow interpretation of this rule, guaranteed that the very possibility of a domain of psychological reality beyond the domain of empirical regularities would never be considered. But there was something supremely ironical about this outcome. For in its flight from a domain of reality stigmatized as "metaphysical," the discipline had limited itself to a kind of reality that turned out to be peculiarly vulnerable to the corrosive effect of social constructivism. The kinds of empirical regularity that were the product of psychological investigation owed their existence to the investigative practices of psychologists. If these regularities also reflected a domain of reality that was independent of these practices, such a domain would have to be defined in terms clearly distinct from the constructed empirical level.

Not that models of psychic causality were anything else but the products of a constructive process, a theoretical rather than a practical construction. However, their existence makes possible the confrontation and reciprocal criticism of theoretically constructed and practically constructed domains. This confrontation offers a possibility of escape from the closed world of unreflected investigative practices but one that can be realized only if the process of criticism really becomes mutual. If traditional investigative practices are considered inviolate, so that only the one-sided criticism of theoretical constructions by the products of practical construction is regarded as legitimate, no progress toward a more reality-adequate kind of knowledge is to be expected.[39] The adequacy of investigative practices must also be measured against the requirements of divergent models of psychological reality.[40] Modern psychology arose in part out of a justified criticism of contrived speculations that were never measured against any systematically constituted empirical domain. Unfortunately, this legacy helped to promote a certain blindness toward an equally unsatisfactory possibility, namely, a discipline based on faith in a limited set of procedures whose fundamental structure is never seriously tested against alternative conceptions of psychological reality.[41]

For most of the twentieth century two circumstances made it difficult for psychological investigation to escape from the web of its own constructions. First, the "cult of empiricism" and its Humean view of reality exerted an iron grip on the discipline. Because empirical domains were necessarily special constructions, theoretical concepts that remained completely tied to them provided no access to anything but a constructed reality. Second, mainstream psychology was isolated from potentially liberating influences. Within the discipline alternative conceptions of theory, practice, and their interrelationship were successfully marginalized. At the same time, the boundaries of the discipline were jealously guarded, thus minimizing the possible impact of alternative models of psychological reality and scientific practice emanating from other social and human sciences and from philosophy. Far from opening themselves to such influences, American psychologists were able to exploit the special congruence of their founding myth with the "American ideal" to assert imperialistic claims over many other areas of human knowledge.[42] This approach did not promote critical reflection about the basis for such claims and allowed the question of access to psychological reality to be decided on a purely technical level.

If, however, we think of reality as a domain that exists independently of empirical constructions, then the question of access to such a domain cannot be decided on the level of a particular empirical investigation. That would imply an epistemic individualism according to which knowledge is the product of an interaction between an individual investigator and nature. But epistemic access to the world is collective[43] – it is always mediated by the social conditions under which groups of investigators work. Moreover, in psychology these conditions are relevant not just for the investigators but also for those who function as the objects of investigation. Like all social work, this collective enterprise is governed by definite rules that are reflected in the form of the product. The knowledge product never consists of a collection of independent elements but always takes the form of an ordered array of some kind. The nature of such an array depends on the ordering principles that are incorporated in the constituting practices.[44]

Moreover, it will have become clear that investigative practices do much more than order observations of a world that is given – they actually prepare the world that is there to be observed. The observations that are available for ordering may function as raw material within the limited framework of a particular investigation, but they are far from raw in terms of any broader perspective. They are made on human subjects who have been actively selected on certain social criteria, who have entered into particular social relations with the investigators, and whose actions have been carefully circumscribed by the investigative situation. Ordering observations once they have been made is only a small part of investigative practice; the major part involves the construction of special "forms of life" in which the observations are grounded.[45] In other words, the work of constituting knowledge domains does not take place only on the level of cognition but involves significant construction on the level of human action and social relations. Knowledge products and the experimental forms of life that generate them are intimately linked. Particular experimental actions and relationships only exist in order to produce knowledge of a certain type, and that knowledge cannot escape the imprint of the forms of life in which it originated.

All this does not mean that we should now replace the naive naturalism of the past with a simpleminded sociological reductionism. To say that psychological knowledge bears the mark of the social conditions under which it was produced is not the same as saying that it is *nothing but* a reflection of these conditions. However, we certainly will not discover what more it might be unless we take the social embeddedness of the knowledge product fully into account when raising epistemological questions. The fundamental mistake that a social constructionist approach can help to correct is embodied in the pervasive assumption that the categories we use in the course of empirical investigation correspond directly to the "natural kinds" that exist in a real world outside the framework of our investigative and intellectual practices.[46] But it does not follow from this that one set of categories and practices is as good as another – that "anything goes." There are defensible criteria for assessing the cognitive yield from the employment of any set of would-be scientific categories and practices. For example, we can make justifiable comparisons among alternative explanatory schemes in terms of their attribute of "depth," and we can assess the social and moral consequences of different kinds of knowledge claims.[47] So far from undermining such comparisons, insights into the historical and socially constructed character of knowledge products are a necessary preparation for making those comparisons.

Of course, in the day-to-day business of technical research such questions do not need to be raised. On this level *specific* knowledge claims can be compared with each other as long as one takes for granted the vast assembly of hidden practical and theoretical assumptions that they share. In other words, the propositional *yield* from the knowledge producing process can be separated from the process itself. But this only results in judgments that are relative to the social framework within which knowledge claims are produced. If we want to raise the question of whether such claims have any validity (or any real meaning) outside this framework, we have to extend our query to the framework itself. This, however, cannot be done effectively without introducing a historical perspective, for such frameworks are historically constituted.[48] Only when we have gained some insight into the kind of historically situated reality to which a received framework has been tied can we raise the question of whether it was ever able – or will ever be able – to transcend that reality.

Addressing that type of question would take us far beyond the limited historical scope of this book. It may however be appropriate to conclude with a few hints in that direction. For example, historical analysis suggests that in the case of psychology any quantitative criterion of reality adequacy is likely to take us in the wrong direction. Thus we cannot attach any epistemic significance to the number of successfully solved empirical problems in a given domain. The one thing that modern research practices in psychology are able to do reliably is to multiply ad infinitum the number of "variables" to be investigated and hence the number of empirical studies and "solved" empirical problems. This has everything to do with the sociology and politics of research and nothing to do with progress in explicating psychological reality.[49]

A more promising procedure for addressing the question of epistemic access is likely to involve the mutual confrontation of divergent empirical domains and the different investigative practices that constitute them. This would have to include domains and practices that, for entirely extraneous reasons, have become identified with other disciplines, like linguistics, sociology, or anthropology.

However, as long as such confrontations are conducted purely as intellectual exercises they are likely to remain without practical consequences. One thing that has emerged from the historical analysis of changes in investigative practices is the importance of the social alliances formed by the discipline as a whole and by subgroups within the discipline. These alliances tended to favor and to maintain certain practices over others. An escape from methodological solipsism is likely to depend on the variety of alliances that members of the discipline manage to forge.

Working relationships and alliances are formed not only with other professional groups. We have seen that the social contexts in which psychological knowledge products are ultimately applied have an effect on the kind of knowledge product for which there is a demand and hence on the practices that must be used to produce it. In that connection it is difficult to ignore the dominant role administratively useful knowledge has played in the past. As long as that state of affairs persisted, the kind of reality to which much of psychological investigation provided access was an administratively created reality. Even when it was not directly tied to actual administrative reality, this kind of research created its own replica of such a reality, as in early American personality research.

The administrative context of application cast its shadow over significant parts of the context of investigation, which did not help to broaden the latter's access to the real world outside such contexts. The prospects of that happening would seem to depend on the extension of disciplinary alliances to groups of people who are more interested in psychological knowledge as a possible factor in their own emancipation than as a factor in their management and control of others.

The worldly success of modern psychology was built on a narrow social basis. That entailed a very considerable narrowing of epistemic access to the variety of psychological realities. Critical analysis can give us some insight into the nature of that narrowing. Further insight depends on some knowledge of that which has been excluded – in other words, knowledge that has emerged in different social contexts. The receptivity of the discipline to such knowledge, however, would seem to be tied to changes in its social and cultural commitments.

Appendix

All but one of the tables that appear in chapters 4, 5, 6, and 8 summarize parts of a content analysis of empirical articles published in psychological journals during the discipline's early years. The selection of journal volumes and articles for inclusion in this analysis was based on the following considerations.

Four basic reference points – 1895, 1910, 1925, and 1935 – were chosen for the purpose of establishing time trends. Journal volumes published during those years were included in the analysis as were the immediately preceding and the immediately following volume. For each set of three consecutive volumes of a journal, the results of the analysis were pooled to allow for minor fluctuations from one year to the next. This yielded data for the periods 1894–1896, 1909–1911, 1924–1926, and 1934–1936. There were minor discrepancies, as when the results of the analysis of the first three volumes of a journal that began publication in 1910 were included in the 1909–1911 period.

Only one of the journals analyzed, the *American Journal of Psychology,* provided a source of empirical articles over the entire period. The other journals began publication at a later stage or ceased publishing empirical studies in significant numbers at some point. The second decade of the twentieth century was a particularly fertile one in terms of the appearance of new American journals. For tracing changes in these journals the 1910 reference point was therefore impractical. Interruption of publication around 1918 introduced a further complication in some cases. To deal with these problems two procedures were adopted. For journals starting publication in this decade the first three volumes were the ones analyzed, irrespective of year of publication. For older journals the analysis of the 1909–1911 period was supplemented by an analysis of volumes published between 1914 and 1916 to reduce the time gap between volumes of these